

BANKING FRAUD DETECTION REPORT

1. Problem Definition:

The objective of this project is to analyse a real world banking transactions dataset to identify suspicious transaction patterns, detect potential fraudulent activities, and generate actionable insights to minimise financial risk.

2. Dataset Description and Source:

The dataset consists of 200,000 bank transaction records. It contains 24 distinct features, including customer details (Age, Gender, State, City), transaction related columns (Transaction_Amount, Transaction_Type, Device_Type, Transaction_Hour), and a binary target variable (Is_Fraud where 1 indicates fraud and 0 indicates a legitimate transaction).

Source: <https://www.kaggle.com/code/marusagar/bank-transaction-fraud-detection-accuracy-95/input>

3. Data Processing Methodology:

Data preprocessing and exploratory data analysis (EDA) were done using Python (Pandas). The methodology included:

- **Data Cleaning:** The dataset was checked for missing values and duplicates, ensuring 100% data integrity.
- **Privacy Compliance:** Personally Identifiable Information (PII), such as Customer_Name and Customer_Email, was dropped from the dataset to ensure data privacy and security.
- **Feature Engineering:** Date and time columns were converted into datetime objects to extract specific features like Transaction_Hour and Day_of_Week to analyze time-based fraud patterns. The cleaned dataset was exported as a CSV file for database import.

4. SQL Design and Query Explanation:

The cleaned data was imported into a MySQL relational database table named banktransactions. SQL was utilized to aggregate and extract key risk metrics.

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- **Schema Design:** A flat-table schema was created with appropriate data types (e.g., INT, DECIMAL, VARCHAR) to store the 200,000 records.
- **Query 1 (Location Risk):** Used GROUP BY State and ORDER BY to identify the top states with the highest volume of fraudulent transactions.
- **Query 2 (Behavioral Risk):** Used GROUP BY Transaction_Type to determine which transaction methods are most frequently exploited by fraudsters.
- **Query 3 (Behavioral Risk):** Used GROUP BY Transaction_Device to determine which transaction devices are most frequently exploited by fraudsters.
- **Query 4 (Behavioral Risk):** Used GROUP BY Day_of_Week to determine on which Day the fraud cases were frequently occurring.
- **Query 5 (Financial Impact):** Utilized conditional logic (CASE WHEN) and aggregate functions (AVG(), MAX()) to compare the average transaction amount of legitimate users versus fraudulent users.

5. Fraud Detection Logic or Model :

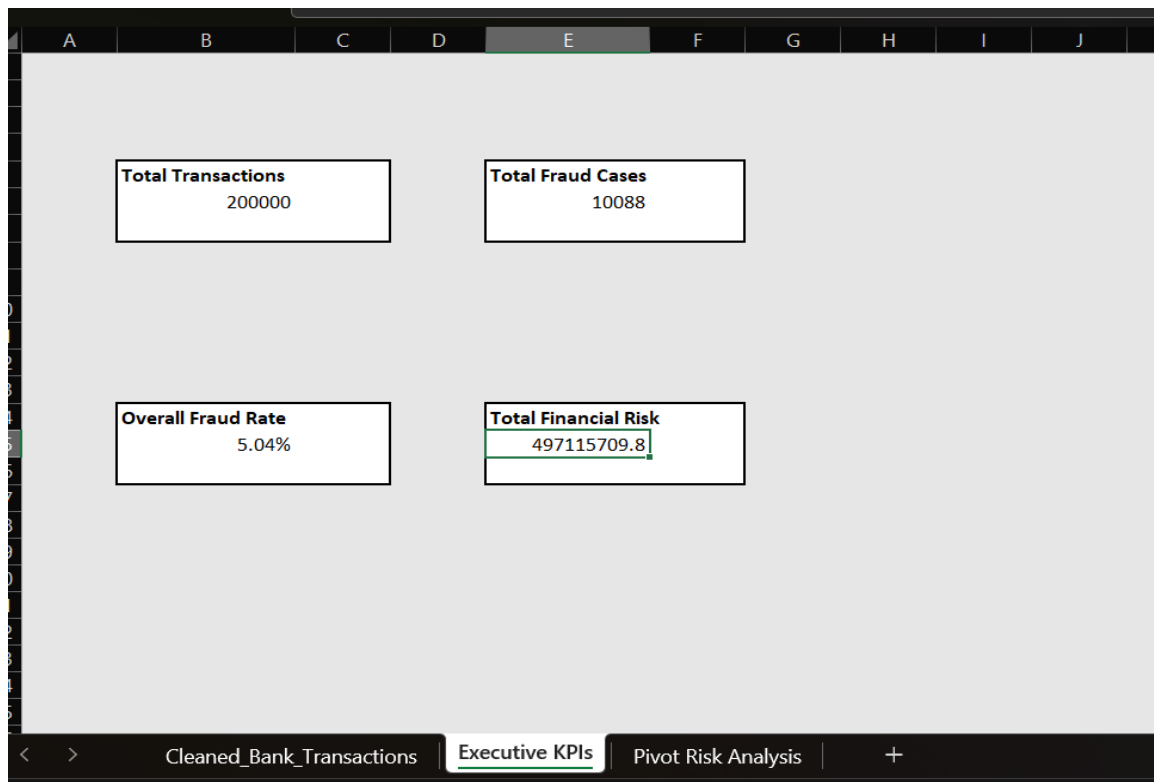
As this project focused on exploratory data analysis and visual risk reporting .The fraud detection logic is **Rule-Based**. Transactions are flagged for review if they cross specific behavioral thresholds identified in the data, such as abnormal transaction hours, specific high-risk transaction types, transaction_devices or high-risk geographical locations.

6. Key Findings and Risk Insights :

Based on the Power BI and Excel pivot analyses, the following key metrics were identified:

- **Overall Risk:** Out of 200,000 total transactions, exactly 10,088 were identified as fraudulent, resulting in an overall fraud rate of 5.04%.

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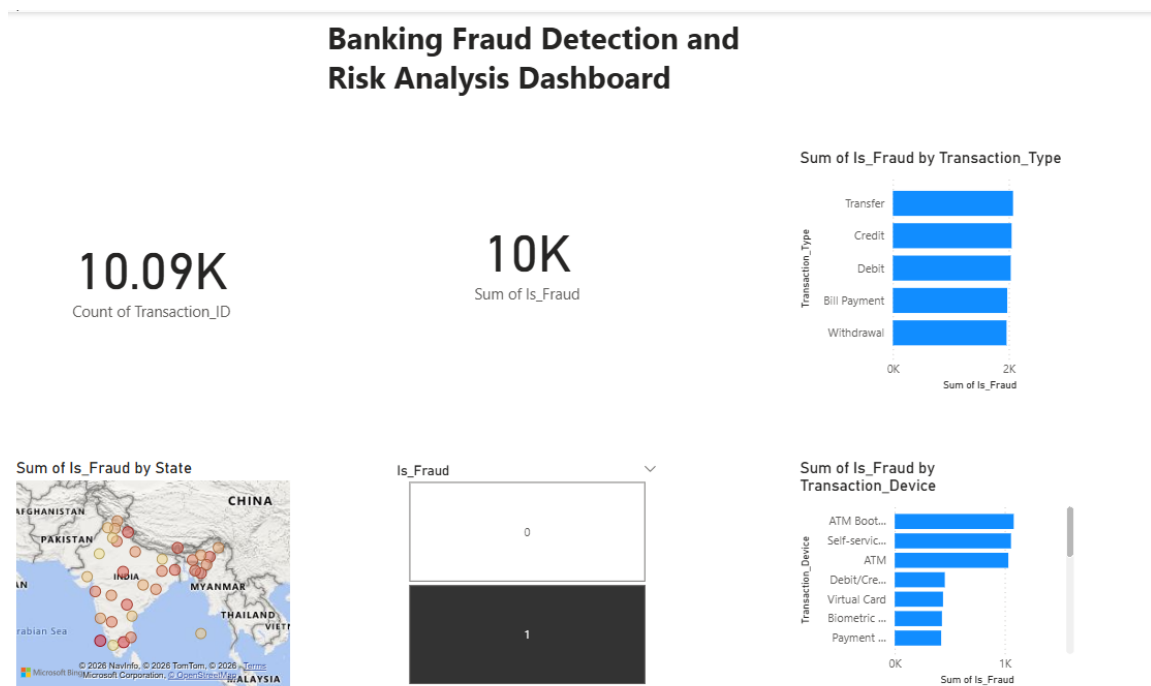


- **Transaction Types:** Analysis revealed the distribution of risk across Credit, Debit, Withdrawals, and Transfers, highlighting which portals require the most security.
- **Device Types:** Analysis revealed the distribution of risk across Desktop, POS, ATM, Mobile highlighting which portals require the most security.

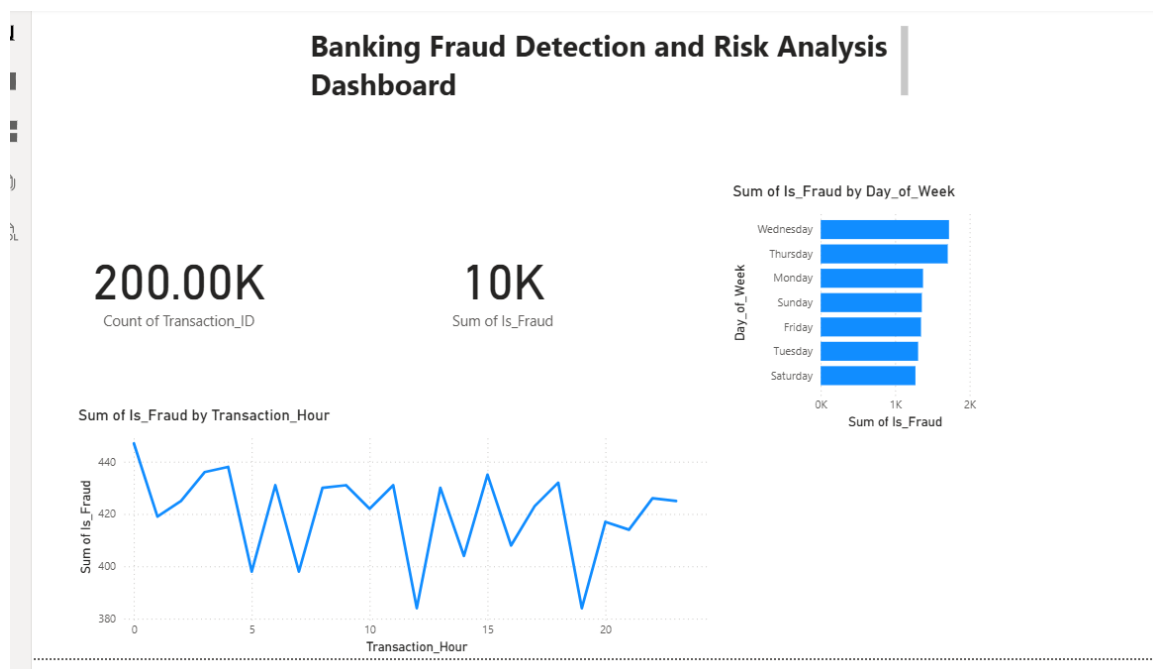
State	Sum of Is_Fraud	Transaction Device	Sum of Is_Fraud	Transaction Type	Sum of Is_Fraud	Day Of Week	Sum of Is_Fraud
Lakshadweep	332	ATM Booth Kiosk	1082	Transfer	2073	Wednesday	1723
Tamil Nadu	328	Self-service Banking Machine	1057	Credit	2048	Thursday	1706
Uttarakhand	319	ATM	1033	Debit	2033	Monday	1374
Telangana	315	Debit/Credit Card	455	Bill Payment	1973	Sunday	1358
Madhya Pradesh	315	Virtual Card	439	Withdrawal	1961	Friday	1347
Sikkim	312	Biometric Scanner	429	Grand Total	10088	Tuesday	1308
Nagaland	312	Payment Gateway Device	422			Saturday	1272
Mizoram	312	Voice Assistant	412			Grand Total	10088
West Bengal	311	Desktop/Laptop	411				
Tripura	308	Smart Card	411				
Dadra and Nagar Haveli and Daman and Diu	305	QR Code Scanner	411	Device Type	Sum of Is_Fraud		
Karnataka	303	Web Browser	408	Desktop	2544		
Delhi	303	Tablet	407	POS	2533		
Puducherry	302	POS Mobile Device	406	ATM	2518		
Meghalaya	302	Wearable Device	399	Mobile	2493		
Jharkhand	301	Mobile Device	397	Grand Total	10088		
Manipur	297	POS Mobile App	391				
Odisha	296	Banking Chatbot	378				
Uttar Pradesh	295	Bank Branch	375				
Maharashtra	294	POS Terminal	365				
Himachal Pradesh	290	Grand Total	10088				

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- **Geographical Hotspots:** The map visualization successfully isolated specific states that carry a higher volume of fraudulent activity.



- **Time-Based Patterns:** Fraud cases were tracked across a 24-hour cycle to visualize risk volume from morning to midnight.



7. Business Recommendations : Based on the identified patterns, the bank should implement the following security measures:

- **Risk-Based OTP (One-Time Password) Verification:** Instead of requiring an OTP for every small, routine purchase, the bank's automated system should trigger an OTP requirement specifically when a transaction matches high-risk criteria. This includes transactions occurring during abnormal hours (e.g., late at night), transactions originating from high-fraud states, or unusually high transfer amounts.
- **Device Verification:** Require secondary authentication whenever an account is accessed from an unrecognized or newly vulnerable Device_Type (e.g., a sudden switch from Mobile to Desktop).

8. Conclusion and Limitations : The analysis successfully separated key variables that correlate with fraudulent banking activity, providing the bank with a clear framework for implementing risk-based security triggers.

- **Data Limitations:** It is important to note that the dataset analyzed was taken from kaggle may be synthetically generated rather than organic real-world data. This is evidenced by the unusually even distribution of the 10,000 fraud cases across all 24 hours of the day and across various states. Real-world fraud typically exhibits severe clustering (e.g., massive spikes between 2:00 AM and 4:00 AM or concentrated entirely in specific cities). While the analytical methodology remains completely valid, real-world deployment of these rules would require adjusting the specific thresholds to match customer transaction patterns.